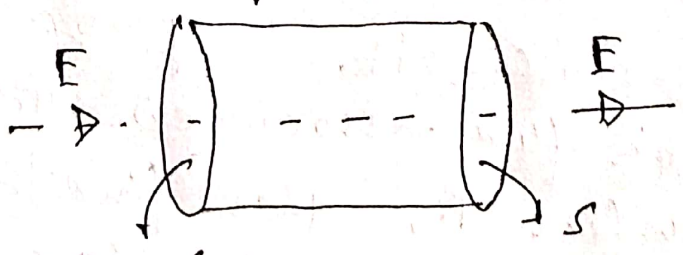
Electric flux, Gauss law.

(1) An electric field $\vec{E} = (4\hat{j} + 3\hat{k})$ N/C is present in a region. A square plate of side 2m is placed parallel to (i) y-z plane (ii) x-z plane. Find electric flux through the plate in each case. [0, 16 V-m]

(2) An electric field is $\vec{E} = 20\hat{i} + 30\hat{j} - 40\hat{k}$ V/m. Find flux of this electric field through a plane of area $0.25\,m^2$ placed
(i) perpendicular to x axis. (5 V-m)
(ii) perpendicular to y axis. (7.5 V-m)

(3) A cylinder of crosssection 'S' is placed in a uniform electric field 'E'. The electric field is along the axis of the cylinder.



Find net electric flux through the cylinder. [0]

(4) Electric flux 'ϕ_1' is entering and flux ϕ_2 is leaving a closed surface. Find total charge enclosed by the surface [$\epsilon_0(\phi_2 - \phi_1)$]

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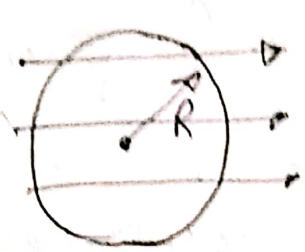
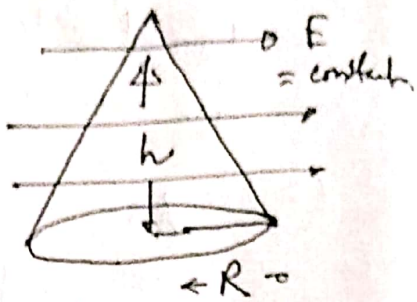
(5) 'n' no of electric dipoles are placed inside a sphere of radius 'r'. Each dipole has charge 'q' & dipole length l. Find net electric flux through the sphere. [Ans zero]

(6)

(a)

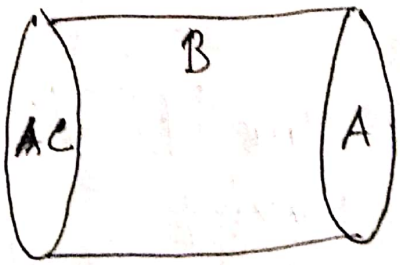


(b)



In fig (a) a sphere, (b) a cone is placed in a uniform electric field E. Find electric flux entering the surfaces in each case [$\pi R^2 E$, hRE]

(7) A hollow cylinder has a charge 'q' Coulomb within it. If 'ϕ' be the electric flux associated with the curved surface B, then find the electric flux linked with plane surface A.



[Ans: $\phi_A = \frac{1}{2}\left(\frac{q}{\epsilon_0} - \phi\right)$]

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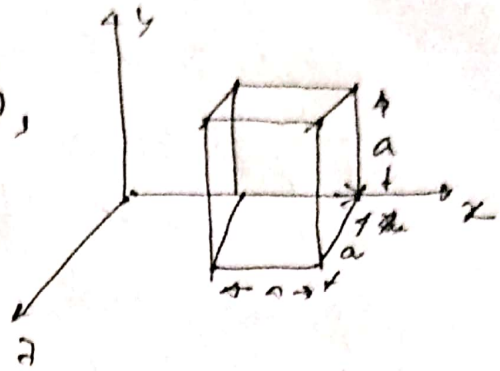
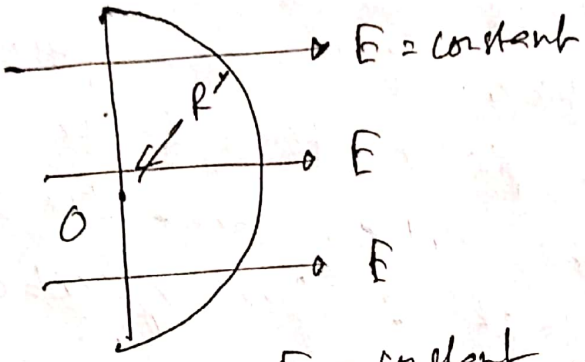
(8) The electric field components in fig are $E_x = \alpha x^{1/2}$, $E_y = 0$, $E_z = 0$,



Calculate flux through the cube of side 'a'

[Ans: $\alpha a^{5/2}(\sqrt{2}-1)$]

(9)



Find electric flux through the curved surface of the hemisphere.

E = constant electric field

R = radius.

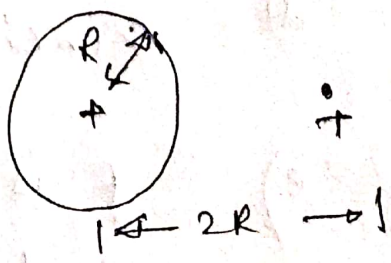
[$\pi R^2 E$]

(10) Find flux of electric field $\vec{E} = 3\hat{i} + 2\hat{j} + \hat{k}$ V/m through a plane of area $50\,m^2$ in (i) x-y plane (ii) y-z plane

[50 V-m, 150 V-m]

(11) Find the flux of electric field through a spherical surface of radius 'R' due to a charge of 10^{-7} C at the centre & and another equal charge at a point 2R away from the centre (fig)

(Ans $1.1 \times 10^4 \frac{N\text{-}m^2}{C}$)



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(12) A charge Q is uniformly distributed over a rod of length l. Consider a hypothetical cube of edge l with the centre of the cube at one end of the rod. Find the minimum possible flux of the electric field through the entire surface of the cube. [Ans: $\frac{Q}{2\epsilon_0}$]

(13) A charge Q is placed at the centre of a cube. Find the flux of the electric field through the six surfaces of the cube. What will be flux through each surface of cube? [$\frac{q}{\epsilon_0}$, $\frac{q}{6\epsilon_0}$]

(14) True OR False:

a) Electric flux $= 0$, implies electric field $= 0$

b) Electric field is zero everywhere, on a closed surface, implies total flux $= 0$

c) $q_{inside} = 0$ implies $\vec{E} = \vec{0}$

d) $q_{inside} = 0$ implies electric flux $\phi = 0$

e) $\vec{E}$ is $\vec{0}$ everywhere on closed surface implies $q_{in} = 0$

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